

2.37 Suppose that $\phi \equiv \phi'$ and $\psi \equiv \psi'$. Show each of the following.

- (a) $\neg\phi \equiv \neg\phi'$
- (b) $(\phi \wedge \psi) \equiv (\phi' \wedge \psi')$
- (c) $(\phi \vee \psi) \equiv (\phi' \vee \psi')$
- (d) $(\phi \rightarrow \psi) \equiv (\phi' \rightarrow \psi')$
- (e) $(\phi \iff \psi) \equiv (\phi' \iff \psi')$

Proof of (a). Let v be any truth assignment.

Suppose $v(\neg\phi) = F$, so that $v(\phi) = T$. Since $\phi \equiv \phi'$, $v(\phi) = T = v(\phi')$ so that $v(\neg\phi') = F$. Similarly, we can show that if $v(\neg\phi') = F$, then $v(\neg\phi) = F$. Thus for all truth assignments v , $v(\neg\phi) = F$ if and only if $v(\neg\phi') = F$, so that $\neg\phi \equiv \neg\phi'$. $\square$

Proof of (b). Let v be any truth assignment.

Suppose $v((\phi \wedge \psi)) = T$, so that $v(\phi) = v(\psi) = T$. We then have $v(\phi') = v(\phi) = T$ as $(\phi \equiv \phi')$ and $v(\psi') = v(\psi) = T$ as $(\psi \equiv \psi')$, so that $v((\phi' \wedge \psi')) = T$. Similarly we can show that if $v((\phi' \wedge \psi')) = T$, then $v((\phi \wedge \psi)) = T$. Thus for all truth assignments v , $v((\phi \wedge \psi)) = T$ if and only if $v((\phi' \wedge \psi')) = T$, so that $(\phi \wedge \psi) \equiv (\phi' \wedge \psi')$. $\square$

Proof of (c). Let v be any truth assignment.

Suppose $v((\phi \vee \psi)) = F$, so that $v(\phi) = v(\psi) = F$. We then have $v(\phi') = v(\phi) = F$ as $(\phi \equiv \phi')$ and $v(\psi') = v(\psi) = F$ as $(\psi \equiv \psi')$, so that $v((\phi' \vee \psi')) = F$. Similarly we can show that if $v((\phi' \vee \psi')) = F$, then $v((\phi \vee \psi)) = F$. Thus for all truth assignments v , $v((\phi \vee \psi)) = F$ if and only if $v((\phi' \vee \psi')) = F$, so that $(\phi \vee \psi) \equiv (\phi' \vee \psi')$. $\square$

Proof of (d). Let v be any truth assignment.

Suppose $v((\phi \rightarrow \psi)) = F$, so that $v(\phi) = T$ and $v(\psi) = F$. We then have $v(\phi') = v(\phi) = T$ as $(\phi \equiv \phi')$ and $v(\psi') = v(\psi) = F$ as $(\psi \equiv \psi')$, so that $v((\phi' \rightarrow \psi')) = F$. Similarly we can show that if $v((\phi' \rightarrow \psi')) = F$, then $v((\phi \rightarrow \psi)) = F$. Thus for all truth assignments, v , $v((\phi \rightarrow \psi)) = F$ if and only if $v((\phi' \rightarrow \psi')) = F$, so that $(\phi \rightarrow \psi) \equiv (\phi' \rightarrow \psi')$. $\square$

Proof of (e). Let v be any truth assignment.

Suppose $v((\phi \iff \psi)) = F$. Since the truth table of $\iff$ is symmetric, WLOG let $v(\phi) = T$ and $v(\psi) = F$. We then have $v(\phi') = v(\phi) = T$ as $(\phi \equiv \phi')$ and $v(\psi') = v(\psi) = F$ as $(\psi \equiv \psi')$, so that $v((\phi' \iff \psi')) = F$. Similarly we can show that if $v((\phi' \iff \psi')) = F$ then $v((\phi \iff \psi)) = F$. Thus for all truth assignments, v , $v((\phi \iff \psi)) = F$ if and only if $v((\phi' \iff \psi')) = F$, so that $(\phi \iff \psi) \equiv (\phi' \iff \psi')$. $\square$